

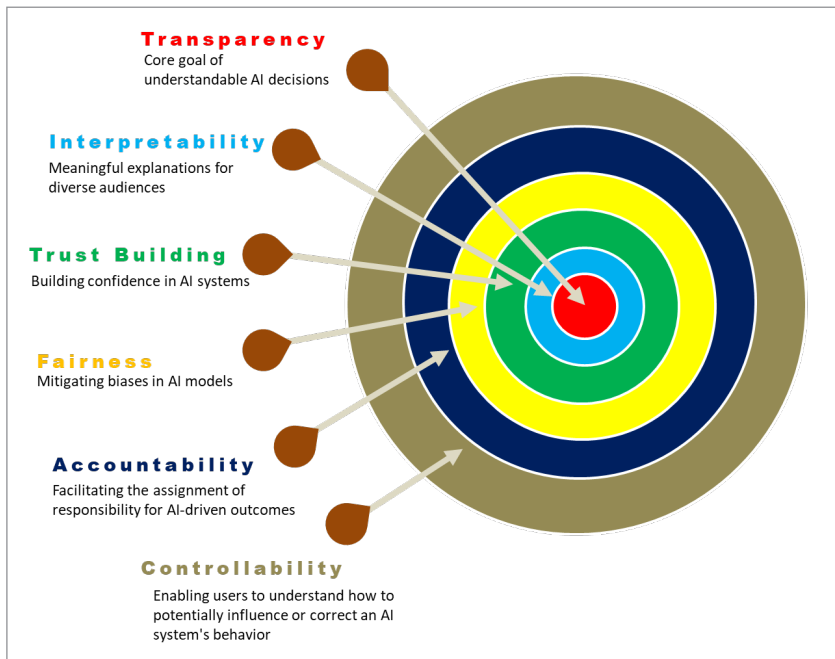


Figure 1: Key objectives of XAI

	Challenge: Why did the vehicle make a specific decision (e.g., lane change, braking)? Critical for accident analysis and safety.
Autonomous Vehicles	XAI Solutions: • Attention maps on sensor data • Rule extraction from control policies
	Challenge: Why did the agent buy/sell in volatile markets? Needed for trust and regulatory compliance.
Intelligent Trading Agents	XAI Solutions: • SHAP: Feature impact analysis • LIME: Local trade explanations
	Challenge: Why did the system suggest a particular diagnosis or treatment? Crucial for patient trust and clinical adoption.
Healthcare Personalized AI Agents	XAI Solutions: • Rule-based diagnostic criteria • Feature importance in treatment plans
	Challenge: Why did the robot choose a specific path or task?
Manufacturing & Logistics Robots	XAI Solutions: • Environment perception visualizations • High-level behaviour rules
	Challenge: Why did the agent propose a hypothesis or experiment?
Scientific Discovery Agents	XAI Solutions: • Logical explanation of proposed steps • data pattern recognition

Figure 2: XAI case studies

their actions becomes critical. The key challenges are:

- **Opaque decision logic:** Most autonomous systems are powered by complex models like deep neural networks, which are ‘black boxes’. It is often extremely difficult to explain why a specific decision was made.
- **Dynamic environments:** Autonomous agents routinely interact with dynamic and unstable environments, so explanations must be more context-dependent and less generalizable.
- **Real-time constraints:** Decisions must be made in milliseconds. Generating explanations that are both helpful and in real-time adds immense additional complexity.
- **Multi-agent interactions:** In systems with large numbers of interacting agents (e.g., swarm robotics or AI for games), an agent’s action may depend on others’ intentions as well.

▪ **User trust and accountability:** Without explanations, users do not trust and developers cannot verify behaviour or meet regulations. Methods such as SHAP, LIME, counterfactual reasoning, and rule extraction assist in solving the puzzle, but complete transparency in agentic AI is still an active research frontier.

The importance of explainable AI for agentic systems becomes even clearer when we examine real-world applications and potential case studies. The case studies in Figure 2 illustrate the application of XAI in a wide range of real-world scenarios. These examples illustrate how XAI techniques can provide crucial insights, build trust, and address ethical concerns in various domains where autonomous agents are being deployed or are on the horizon.

Understanding ‘why’ an agent behaves is not only a theoretical benefit, but a practical necessity in order to achieve the full potential of agentic AI while reducing its dangers.